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 Studierichting: Informatica
 Oefeningenreeks 6G nr. 2.8

$$f(x) = e^{|x|}$$

$$f(-x) = e^{|-x|} = e^{|x|} = f(x)$$

$$\Rightarrow f \text{ is even} \Rightarrow b_n = 0$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx))$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{|x|} dx$$

Omdat f even is:

$$= 2 \cdot \frac{1}{\pi} \int_0^{\pi} e^x dx$$

$$= \frac{2}{\pi} [e^x]_0^{\pi}$$

$$\Rightarrow a_0 = \frac{2}{\pi} (e^{\pi} - 1)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{|x|} \cos(nx) dx$$

Omdat f even is:

$$= \frac{2}{\pi} \int_0^{\pi} e^x \cos(nx) dx$$

$$\begin{cases} u = \cos(nx) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = -n \sin(nx) dx \\ v = e^x \end{cases}$$

$$I = e^x \cos(nx) - n \int_0^{\pi} e^x \sin(nx) dx$$

$$\begin{cases} u = \sin(nx) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = n \cos(nx) dx \\ v = e^x \end{cases}$$

$$J = e^x \sin(nx) - n \int_0^{\pi} e^x \cos(nx) dx$$

$$\Rightarrow I = e^x \cos(nx) - n(e^x \sin(nx) - nI)$$

$$= e^x \cos(nx) - ne^x \sin(nx) + n^2 I$$

$$I(1 + n^2) = e^x \cos(nx) - ne^x \sin(nx)$$

$$= e^x (\cos(nx) + n \sin(nx))$$

$$\begin{aligned}
&= \frac{2}{\pi} \left[\frac{e^x (\cos(nx) + n \sin(nx))}{1+n^2} \right]_0^\pi \\
&= \frac{2}{\pi} \left(\frac{e^\pi (-1)^n}{1+n^2} - \frac{1}{1+n^2} \right) \\
\Rightarrow a_n &= \frac{2(e^\pi (-1)^n - 1)}{\pi(1+n^2)} \\
f(x) &\sim \frac{e^\pi - 1}{\pi} + \sum_{n=1}^{\infty} \frac{2(e^\pi (-1)^n - 1)}{\pi(1+n^2)} \cos(nx)
\end{aligned}$$

References